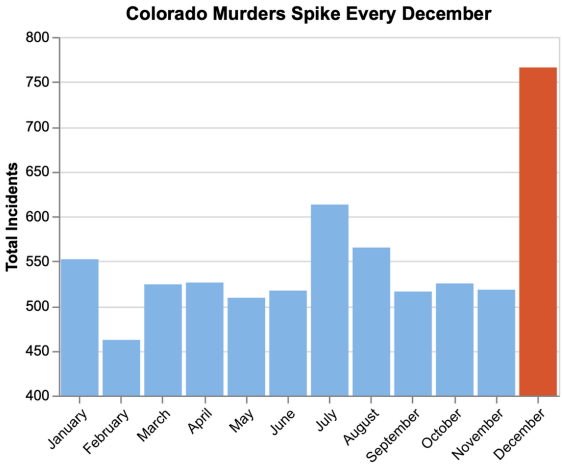
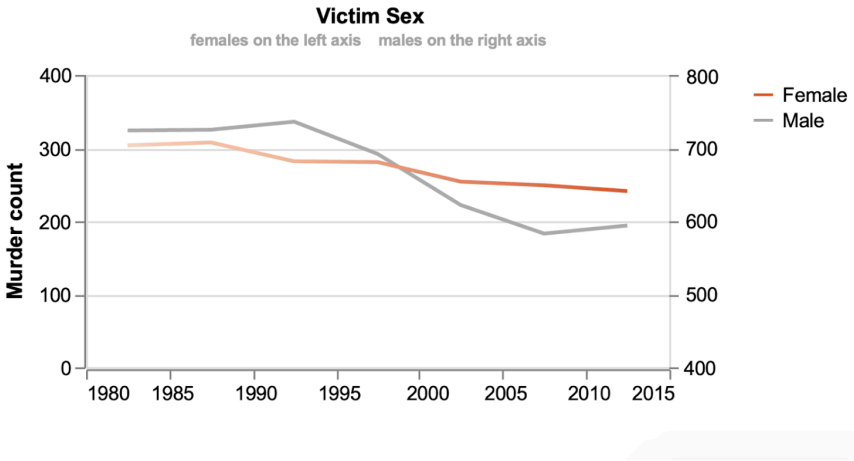
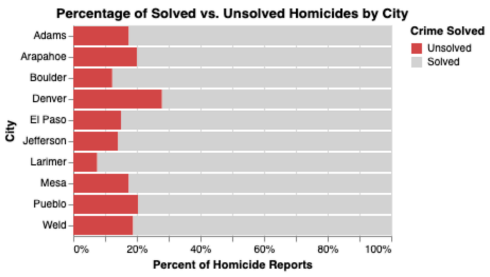
Are Colorado's Murders Really Getting Worse? Depends How You Look at the Data

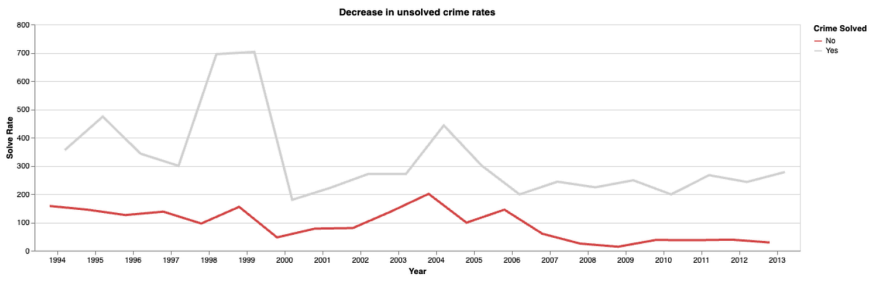
How the exact same crime data can tell two completely opposite stories

Authors: Kaliya Sonoqui, Roxy Houck, Rogan Tinjum, Ben Cotton









Summary

Our project explores the trends hidden in a dataset of U.S. murder records from 1980–2014. The original dataset covered crime across the entire country, but with over 638,000 records, it was far too large to meaningfully explore, so we filtered it down to Colorado only, leaving roughly 6,600 homicide records to work with. Using that single dataset, we built two truthful visualizations showing that

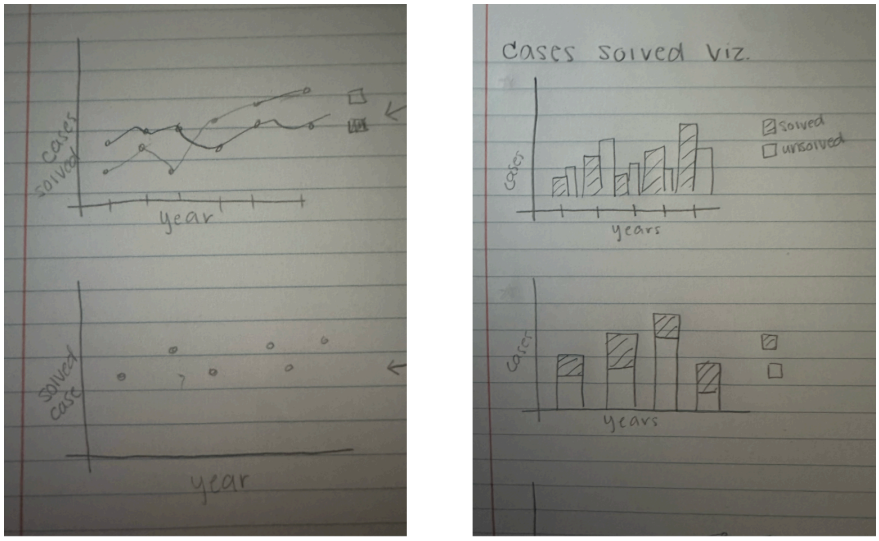
Colorado's unsolved crime rate has been shrinking over time, and two deceptive visualizations using the same real numbers to falsely suggest that murders spike every December and that women are increasingly being targeted.

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Sketching & Prototyping: Truthful

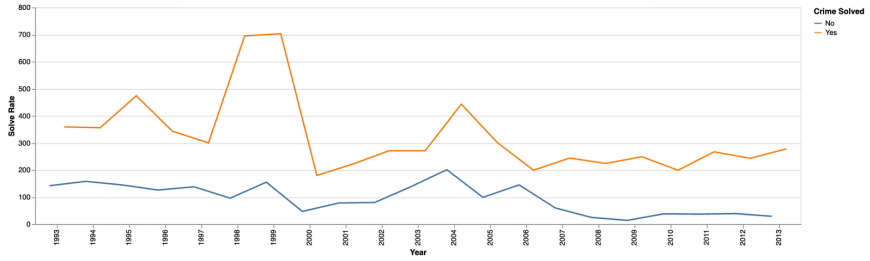
Decrease in Unsolved Crime Rates

In the process of creating the "good" graph showing the decreasing unsolved rates over time, I first started with sketching a few rough drafts on paper in order to see basically what the data could look like in a visualization. Once they were done I showed them to people and briefly described what the story for the visualization was and asked them their opinions about the sketches, documenting the opinions given. At first I thought the best option would be a bar graph with the crime rate on the y axis and years on the x axis. I wanted to do a cluster graph showing the solved vs unsolved rates right next to each other for easy comparison. However after receiving the feedback from my user testers, I changed concepts to the line graph which is a better way to show the relationships in the data alongside the time passing.



First low-fidelity prototyping for First Truthful Graph

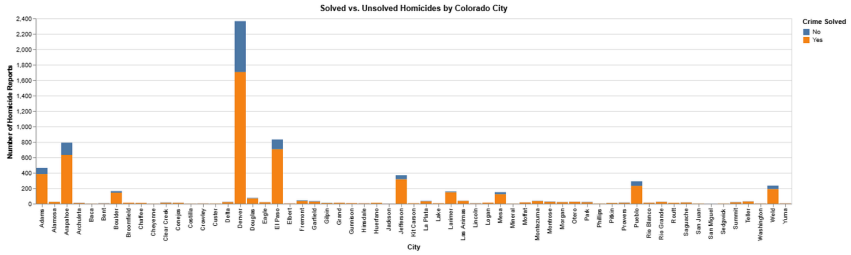
Once I made the decision to make a line graph, I started the design by placing the year on the X axis and the crime solve rate on the Y axis. I added in the axis titles next to make the data clear and easy to read. Next I put the legend in, using color to distinguish between the solved and unsolved data. After I had the general outline for my graph laid out I started making it easier to read and understand. I started by increasing the lines width so it is easier to see. Next I changed the color palette, my first iteration I had green for solved rates and red for unsolved rates. However I found this took away from the story I was trying to tell by drawing the focus away from the line I really wanted people to look at (Unsolved Rates). So in my second iteration I turned the solved line grey, keeping the line for comparison so people could see solved crime rates increasing and drawing more focus to the red line that communicated my story of unsolved rates decreasing in recent years. Next I rotated the year ticks on the x axis to be horizontal instead of vertical to be easier to read. Finally I added a title to the top of the graph to show my story more clearly.



Second prototype for First Truthful Graph

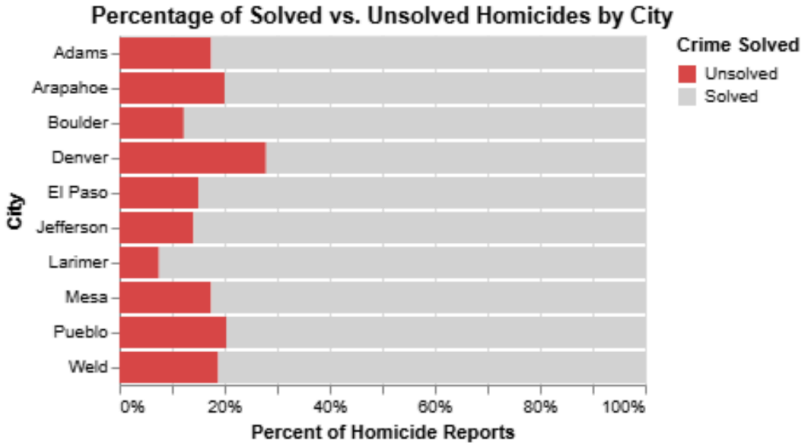
Percentage of Solved vs. Unsolved Homicides by City

We initially compared raw solved/unsolved counts across *all* Colorado cities, but this made larger cities stand out simply because they had more reports, the sheer number of locations also made the chart cluttered and hard to compare.



First Prototype for Second Truthful Graph

To fix this, we narrowed our scope to the ten cities with the most homicide reports and switched from raw counts to percentages, turning the chart into a 100% horizontal stacked bar graph, with red representing unsolved cases and gray representing solved ones.



Final design using percentages for the ten locations with the most homicide reports

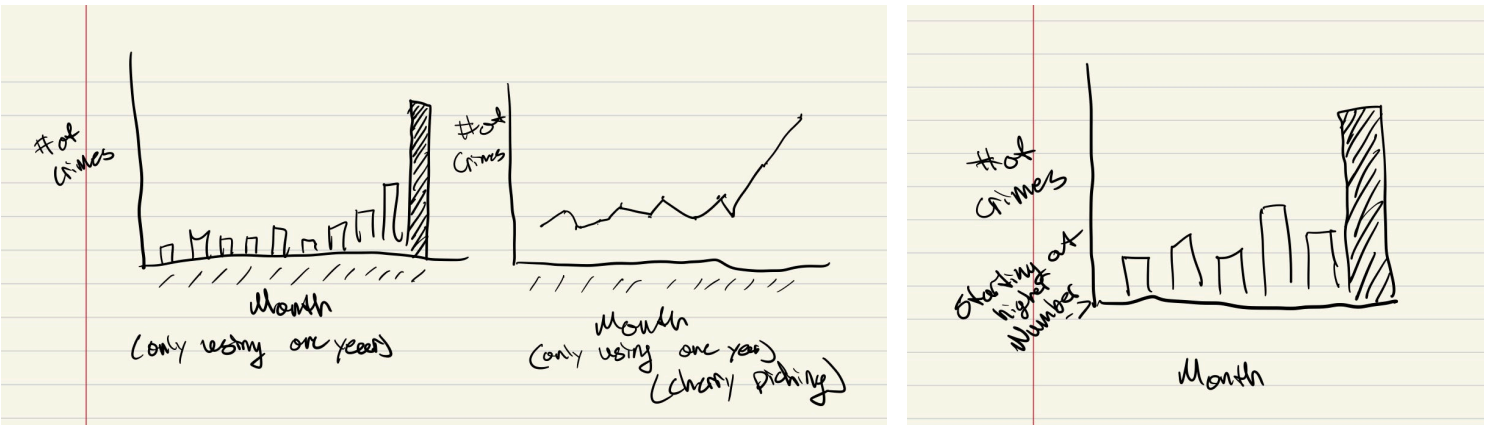
This design choice connects directly to pre-attentive processing—our ability to "immediately recognize variation with little or no conscious effort" (Peck, 04b-Perception-Part2-FA26). Using consistent bar lengths and a clean 0–100% scale makes the proportion of unsolved cases in each city instantly comparable, without a viewer needing to do any mental math.

One strength of this design is that it makes differences in unsolved percentages easy to compare across cities at a glance. One weakness is that using percentages hides the total number of homicide reports in each city—and limiting the graph to only the top ten locations leaves out lower-volume cities entirely. With more time, we'd consider adding total case counts as a tooltip.

Sketching & Prototyping: Misleading

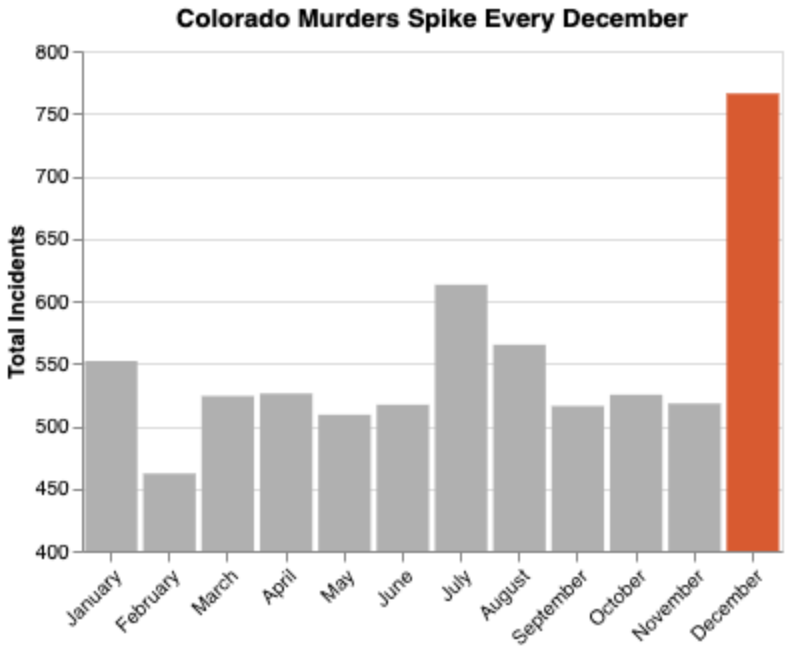
"Colorado Murders Spike Every December"

We started by prototyping ideas that used common misleading-graph techniques, sketching several rough concepts to see which would work best with our data. Our first idea was to isolate a single year where December's crime count happened to be dramatically higher than every other month. After reviewing that sketch, we realized that adding a title would let readers catch on almost immediately that we were only using one year's data. So we pivoted to using *all-time monthly averages* instead, paired with a skewed y-axis, to make the December increase look far more dramatic than it really is.



First and second low-fidelity prototyping for First Misleading Graphs

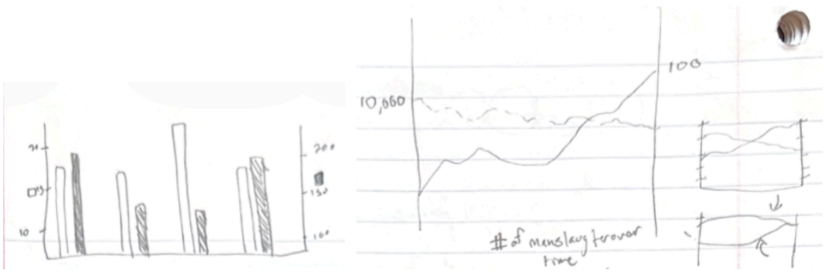
This design breaks a core principle of graphical integrity: the **q-ratio** *—a bar that is 2/3 the height of another should reflect a data-point that is 2/3 of another.* By starting our y-axis at 400 instead of 0, we broke that relationship on purpose. December's bar (766 incidents) is only about **1.60x** the size of February's (462 incidents) in the real data—but because we cropped out everything below 400, December's bar appears roughly **2x-3x** taller than February's on the page. The visual gap is more than three times larger than the real gap, which is exactly what makes the "spike" look so much more dramatic than it actually is.



Final Design of the Misleading Chart "Colorado Murders Spike Every December"

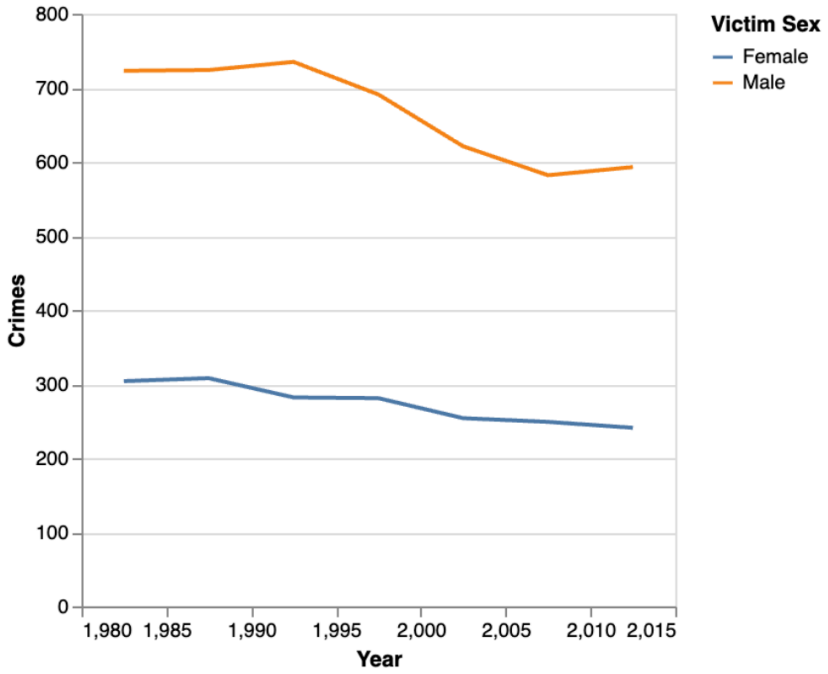
"Modern Women Are Targeted Around the Holidays"

The design that stood out most for our second misleading graph was a dual-axis chart, a classic and effective way to obscure a true comparison.



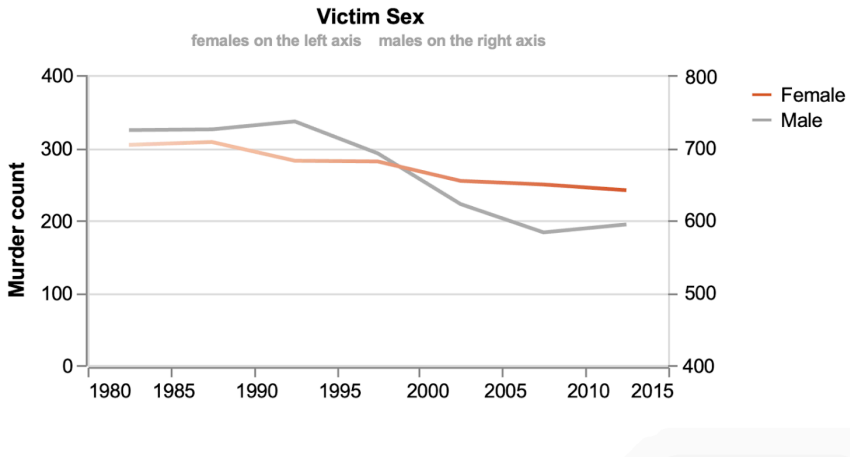
First and second low-fidelity prototyping for Second Misleading Graphs

After sketching a few dual-axis concepts, we decided the real key to the deception would be using color to focus the reader on the *lines* rather than the *axes*. Our first Altair iteration encoded male victims in blue and female victims in orange, but the two colors had similar brightness, so neither line stood out.



Thirs Prototype for Second Misleading Graph

Since we wanted the female trend to be the one readers fixated on, we changed that line to a bright red and faded the male line to gray. This kept both lines technically visible for "comparison," while pulling nearly all visual attention toward the line supporting our false narrative, a direct application of how color can outweigh even size or position in controlling where a viewer's eye goes first.



Final Design of the Second Misleading Chart

User Testing

Truthful Graphs

When we tested a clustered bar graph, a stacked bar graph, and a line graph for the unsolved-rate story, testers told us the clustered bar was the "easiest BAR GRAPH to read and make comparisons with," but that the stacked bar was "harder to see the comparisons on top of each other." The line graph won out as "the easiest to see the comparisons in the data and see the time passing" which is exactly why we ultimately chose it over both bar options.

During our in-class demo day, several people said the city percentage graph was easy to read and compare across locations. One person found the "Yes"/"No" legend labels confusing and suggested "Solved"/"Unsolved" instead—a clear, easy fix we'd want to make in a future version. Another noted the light gray could be difficult for some viewers to see, which is worth revisiting for accessibility.

Misleading Graphs

For the December chart, we asked our roommates for their gut reactions before explaining anything. Across the board, they said it was immediately clear that December had a higher crime rate "by a landslide" confirming the deception worked on a first glance. Once we explained our actual goal, they, being engineers, were able to quickly pick apart *how* each graph was misleading them. Their most useful suggestion was to grey out all the other months even further to make December's bar pop even more, which we implemented in our final version.

For the dual-axis chart, we tested with people who had no prior data-vis background to get an unbiased read. The biggest takeaway: it did take them a moment to realize the data was plotted on two separate y-axes—a bit sooner than we'd have liked, meaning the illusion wasn't as airtight as we hoped. The most common comment was about color: because we had three labeled axes in black text of the same weight as our title, several testers said they looked at the *text* first, before ever looking at the lines, which weakened the intended effect. In response, we faded the axis numbers to a light gray while keeping the axis lines themselves dark, so the chart still felt

"grounded," but the color contrast on our two data lines could do more of the work in guiding the eye.

. . .

Reflection

Our final prototypes make clear how far design choices alone can shift a reader's takeaway: the truthful graphs communicate an honest, easy-to-verify trend of falling unsolved rates, while the misleading graphs use that same exact data to sell a false one, proof that the technique works. That said, the prototypes have real limits. Because we narrowed a national dataset down to homicides in Colorado alone, nothing here generalizes beyond homicide-specific patterns in one state, it can't speak to "crime" more broadly. Our color choices also lean heavily on red-versus-gray to direct attention across nearly every chart, which may not read clearly for colorblind viewers, a limitation that goes beyond the "light gray was hard to see" feedback we got from classmates, who also flagged our "Yes"/"No" legend as confusing and suggested "Solved"/"Unsolved" instead. With more time, we'd want to test with a larger and more varied group of outside reviewers, expand our scope to more states or a longer time range to see if the patterns hold, and rework our color palette to be accessible to colorblind viewers rather than relying so heavily on red and gray.